

Parnika Sarbahi

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Summary

Computer Science student with hands-on experience building autonomous LLM-powered security agents using tool-use and function calling. Built VulnAgent, a production-grade agentic pipeline with code-controlled triage flow, confidence-based human escalation, and a labelled eval harness – diagnosing and fixing real LLM tool-calling failures rather than trusting output blindly. Proficient in Python, REST APIs, Git, and CI/CD, with applied security tooling across Bandit, SAST concepts, Nmap, and Wireshark.

Education

UPES <i>B.Tech in Computer Science (Cyber Security & Digital Forensics)</i>	July 2023 – June 2027 <i>CGPA: 8.91/10.0</i>
Army Public School <i>CBSE Higher Secondary Certificate</i>	March 2021 – May 2023 <i>Percentage: 85.0%</i>
Loreto Convent Intermediate College, Lucknow <i>ICSE Secondary School Certificate</i>	March 2010 – March 2021 <i>Percentage: 95.83%</i>

Technical Skills

AI Agents and LLM Tooling: LLM tool-use and function calling, agentic pipeline design, structured output, confidence-based escalation guardrails, eval framework design, prompt engineering, model-agnostic architecture (Ollama, local LLMs)

Languages: Python, Java, Bash scripting

Security Tooling: Bandit (SAST), vulnerability triage and remediation, Nmap, Wireshark, Burp Suite, log analysis, network intrusion detection

Security Concepts: Secure coding, access control, IDS/IPS, firewall configuration, PKI, endpoint security, CIA triad, vulnerability assessment

Engineering: REST APIs, Git, GitHub Actions (CI/CD), pydantic, modular architecture, OOP, design patterns

AI/ML: PyTorch, TensorFlow/Keras, CNN-BiLSTM, Transformer, SHAP, LIME

Tools and Systems: Linux, MySQL, multithreading

Experience

Cybersecurity and Software Development Intern <i>AERSPL (Abhima Energy Recovery System Pvt. Ltd.)</i>	May – July 2025 <i>Remote</i>
- Eliminated recurring data-reliability failures on a DST-funded industrial automation project by replacing fragile ad-hoc scripts with configuration-managed Python modules for structured MySQL data management. - Reduced attack surface of a cyber-physical robotic system by auditing the Python codebase and enforcing input validation, credential management, and access control across all modules.	

Projects

VulnAgent: Autonomous LLM Vulnerability Triage Agent <i>Python, Ollama, Bandit, pydantic, GitHub Actions</i>
- Eliminated manual triage for Bandit SAST findings by building an autonomous agent harness that enforces a deterministic, code-controlled pipeline (classify_severity, suggest_remediation, generate_ticket, or escalate_to_human) via four tool-use calls – Python guarantees the sequence, the LLM supplies the reasoning within each forced step. - Diagnosed and resolved three production-grade LLM failure modes: redesigned control flow from model-driven to code-driven after the local model would classify then stop; fixed a confidence-elicitation failure where a 0.0–1.0 float degenerated to zero by switching to LOW/MEDIUM/HIGH categorical labels; root-caused a 100% silent tool-failure rate via a custom diagnostic script and built a resilient fallback parser to recover malformed structured output. - Validated reliability against a 20-finding hand-labelled eval set – 85% within-one-severity accuracy, 95% escalation accuracy, 100% recall on CRITICAL findings – and integrated into a live GitHub Actions CI/CD pipeline that auto-fails builds on high-severity findings; every failure mode has a logged fallback so no finding is silently dropped.
Explainable AI-Powered IoT Intrusion Detection System <i>Python, TensorFlow, CNN-BiLSTM, SHAP, LIME, Flask</i>
- Trained a CNN-BiLSTM IDS model on 158,964 balanced IoT-23 samples (1.4M rows, 17.8x imbalance corrected) on NVIDIA H100, achieving 97% accuracy and AUC-ROC 0.9964 across DDoS, port scan, botnet, and C&C classifications. - Designed SHAP and LIME evaluation pipelines to validate model output trustworthiness – surfacing which network features drove each verdict – and deployed a Flask application with real-time .csv/.pcap upload and interpretability reporting.

JScanSec v2.0 – Network Security and Forensics Toolkit | *Java, MySQL, Multithreading, TCP/IP*

- Architected a 10-module security toolkit in Java (Strategy Pattern) spanning TCP/IP port scanning, DNS recon, SSL/TLS certificate analysis, OWASP-aligned web vulnerability detection, and authentication testing – each module independently pluggable; engineered 50-thread port scanner achieving 60x throughput improvement with unit tests and MySQL audit logging.

Hybrid CNN–Transformer Image Watermarking System | *Python, PyTorch, CNN, Transformer*

- Independently designed a CNN–Transformer hybrid with DWT-DCT frequency-domain embedding achieving PSNR $\approx$ 52 dB and SSIM $\approx$ 0.999 under 10+ adversarial attacks. **Accepted at IEEE INCRYPOL 2026, JNU New Delhi, October 2026.**

Technical Certifications

Junior Cybersecurity Analyst Career Path – Cisco Networking Academy **June 2026**

- Threat analysis, IDS/IPS, security alerts, firewall and cloud security, network defense, SOC analyst workflows.

Network Defense – Cisco Networking Academy **May 2026**

- Firewall configuration, PKI, identity lifecycle management, endpoint security, secure data transport.

Foundations of Cybersecurity – Google **2024**

- CIA triad, access control, threat classification, Python and SQL in security, security operations.

Oracle Cloud Infrastructure – AI Foundations Associate **2025**

- Core AI, ML, deep learning, and Generative AI with hands-on cloud applicability.

Leadership and Extracurricular

Google Developer Groups (GDG) – UPES **December 2023 – August 2025**

Core Member

Dehradun, Uttarakhand

- Delivered a Python and ML workshop to 80+ students at GDG ML Bootcamp, owning curriculum design, logistics, and technical delivery end-to-end.
- Grew GDG LinkedIn presence by 200+ followers over 1.5 years through consistent technical content and event promotion.